

Project Name :- Amazon Sales Analytics Platform

1. Project Overview

The rapid growth of e-commerce has resulted in the generation of massive volumes of transactional sales data. Organizations collect valuable information related to customer purchases, product performance, payment behavior, pricing, and regional sales every day. Transforming this raw data into actionable business insights enables organizations to make informed decisions, improve operational efficiency, and enhance customer satisfaction.

This project develops an end-to-end AI-Powered Amazon Sales Analytics Platform using Python, SQL, and Power BI. The solution follows a complete data analytics workflow beginning with business understanding, data validation, exploratory data analysis, SQL-based business analysis, and interactive dashboard development.

The project aims to analyze customer purchasing behavior, product performance, sales trends, payment preferences, and regional sales performance while generating meaningful business insights that support strategic decision-making.

2. Business Problem

Modern e-commerce businesses process thousands of customer orders every day across multiple product categories and geographical regions. Without a structured analytics solution, organizations struggle to identify revenue trends, customer purchasing patterns, operational inefficiencies, and product performance.

Decision-makers require timely and accurate insights to optimize pricing strategies, manage inventory effectively, improve marketing campaigns, and enhance customer experience. Manual reporting methods often fail to provide the level of detail needed for proactive business planning.

This project addresses these challenges by transforming raw transactional sales data into meaningful analytical insights that support data-driven business decisions.

3. Business Objectives

The objectives of this project are:

- Perform end-to-end exploratory data analysis using Python.
- Conduct SQL-based business analysis to answer real-world business questions.
- Analyze overall sales performance.
- Identify top-performing products and categories.
- Analyze customer purchasing behavior.
- Identify the most preferred payment methods.
- Measure the impact of discounts on sales.
- Evaluate regional sales performance.
- Identify best-performing brands.
- Monitor order status and cancellations.
- Build an interactive Power BI dashboard.
- Generate AI-assisted business insights.
- Provide actionable business recommendations.

4. Industry Background

The e-commerce industry is highly competitive, requiring organizations to continuously monitor customer behavior, product demand, pricing strategies, and operational performance. Companies increasingly rely on data analytics to understand sales trends, improve customer satisfaction, optimize inventory, and maximize profitability.

Data-driven decision-making enables businesses to identify high-performing products, evaluate promotional effectiveness, forecast demand, and improve supply chain efficiency. Modern analytics platforms allow organizations to move beyond historical reporting by providing interactive dashboards, predictive insights, and real-time performance monitoring.

5. Dataset Overview

The dataset consists of transactional sales records collected from an Amazon-like e-commerce platform. Each record represents an individual customer order and contains information related to customer details, purchased products, pricing, discounts, taxes, payment methods, shipping costs, seller information, and geographical location.

Column	Business Meaning
OrderID	Unique identifier for each order
OrderDate	Date when the customer placed the order
CustomerID	Unique customer identifier
CustomerName	Name of the customer
ProductID	Unique product identifier
ProductName	Name of the product
Category	Product category
Brand	Product brand
Quantity	Number of units purchased
UnitPrice	Price of one unit
Discount	Discount applied
Tax	Tax charged
ShippingCost	Delivery cost
TotalAmount	Final amount paid by customer
PaymentMethod	Customer payment method
OrderStatus	Order status (Delivered, Cancelled, etc.)
City	Customer city
State	Customer state
Country	Customer country
SellerID	Seller identifier

6. Project Scope

- Data Quality Assessment & Validation
- Exploratory Data Analysis (EDA)
- Sales Performance Analysis
- Customer Behavior Analysis
- Product Performance Analysis
- Financial Analysis
- Regional Sales Analysis
- Operational Performance Analysis
- Advanced SQL Analytics
- Interactive Dashboard Development
- KPI Monitoring
- Business Recommendation Generation
- Data-Driven Decision Support

7. Key Business Questions

The following business questions will be answered during the analysis:

1. Sales Analysis

- What is the total revenue generated?
- Which product category generates the highest revenue?
- Which products are sold the most?
- Which brands perform the best?
- What are the monthly sales trends?
- Which products generate the highest sales value?

2. Customer Analysis

- How many unique customers are there?
- Which customers purchase most frequently?
- What is the average order value?
- Which payment methods are most preferred?

3. Regional Analysis

- Which countries generate the highest revenue?
- Which states generate the highest sales?
- Which cities contribute the most revenue?

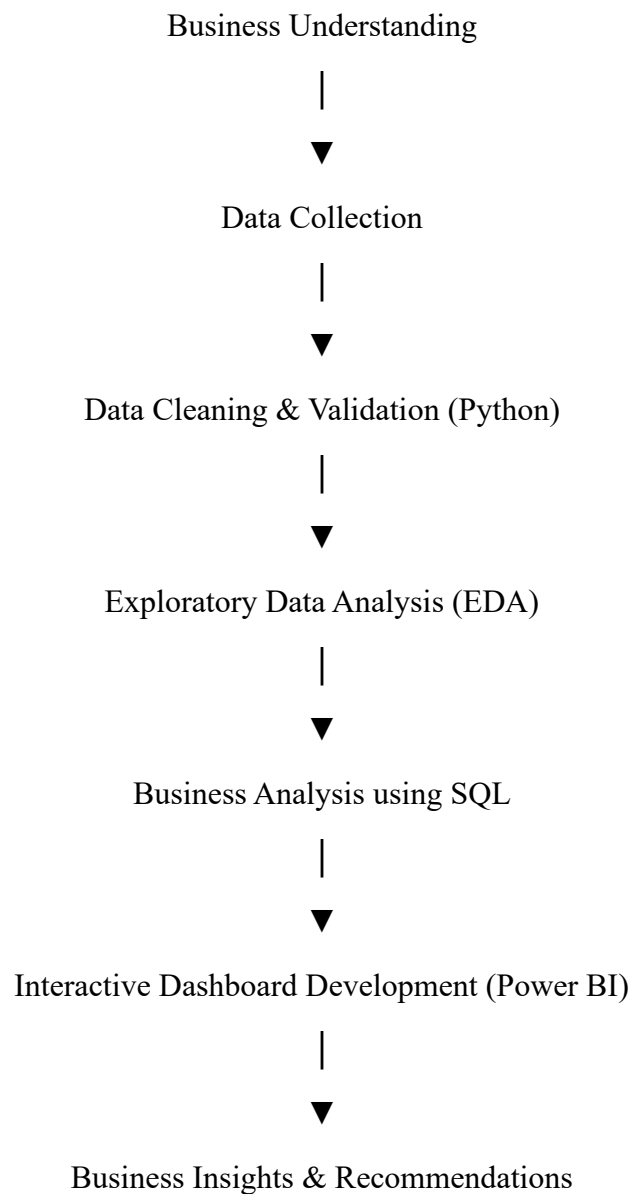
4. Operational Analysis

- How many orders are delivered?
- How many orders are cancelled?
- What is the cancellation rate?
- What is the average shipping cost?
- How much discount is provided overall?

5. Financial Analysis

- Which categories receive the highest discounts?
- Which brands generate maximum revenue?
- What is the average tax collected?
- What is the average revenue per order?

8. Project Overflow



9. Expected Business Outcomes

After completing this project, the business will be able to:

- Identify high-performing products and brands.
- Understand customer purchasing patterns.
- Monitor regional sales performance.
- Improve inventory planning.

- Optimize pricing and discount strategies.
- Improve delivery performance.
- Support data-driven decision-making through interactive dashboards and business reports.

10. Key Performance Indicators (KPIs)

Category	KPIs
Revenue	Total Revenue, Average Order Value, Revenue by Category, Revenue by Brand
Sales	Total Orders, Quantity Sold, Best Selling Product, Best Selling Category
Customer	Total Customers, Repeat Customers, Average Products per Order, Preferred Payment Method
Operations	Delivered Orders, Cancelled Orders, Cancellation Rate, Average Shipping Cost, Average Discount

11. Tools & Technologies

Technology	Purpose
Python	Data Cleaning & EDA
Pandas	Data Manipulation
Matplotlib	Data Visualization
SQL (MySQL)	Business Analysis
Power BI	Dashboard Development
GitHub	Documentation & Version Control

12. Deliverables

Deliverable	Description
Business Understanding Report	Documents the business problem, objectives, project scope, stakeholders, and expected outcomes.
Raw Dataset	Original Amazon sales dataset used for analysis.
Cleaned Dataset	Preprocessed dataset after handling data quality issues.
Python Notebook	Data cleaning, validation, and exploratory data analysis performed using Python.
EDA Report	Visual analysis highlighting sales trends, customer behavior, and product performance.
SQL Database	Structured database containing imported sales records.
SQL Business Analysis	Business-oriented SQL queries answering key analytical questions.
Advanced SQL Analytics	CTEs and Window Functions demonstrating advanced SQL techniques.
Power BI Dashboard	Interactive dashboard for monitoring KPIs and business performance.
Executive Business Insights	Summary of key findings derived from Python, SQL, and Power BI analysis.
Business Recommendations	Actionable recommendations to improve sales, marketing, pricing, and operations.
GitHub Repository	Complete project source code, documentation, datasets, SQL scripts, notebooks, and dashboard files.
README Documentation	Comprehensive project overview, workflow, technologies, and setup instructions.